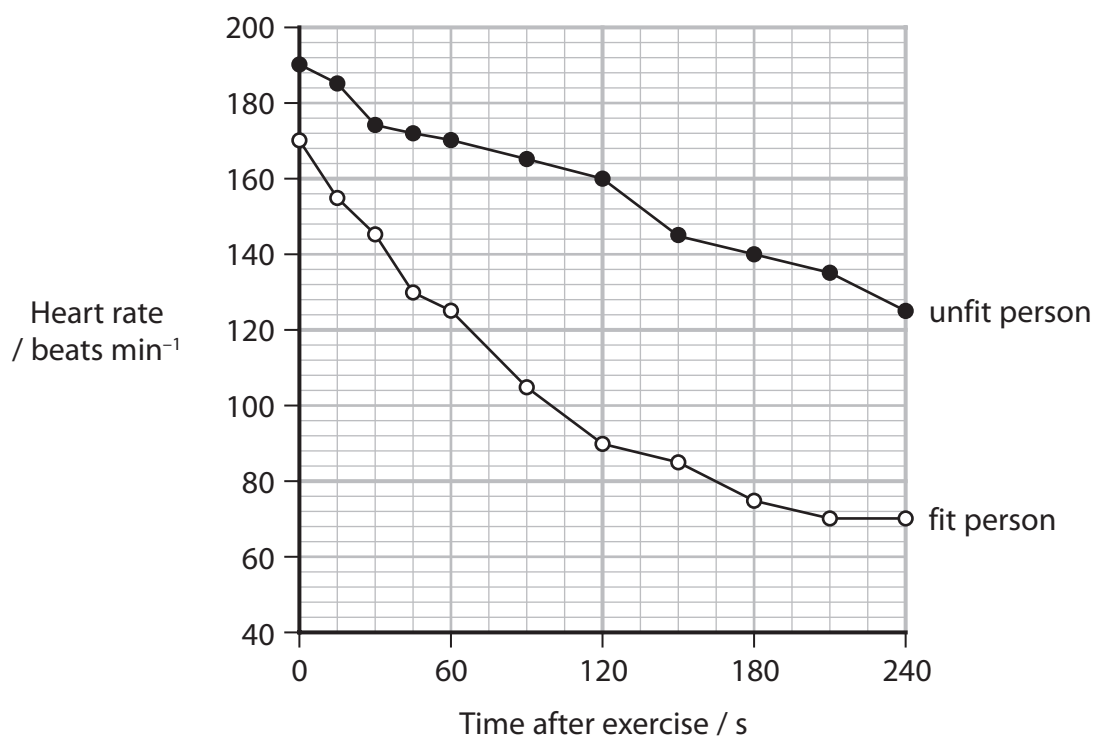


6 Exercise affects the cardiac output of a person.

The graph shows the change in heart rate following exercise for an unfit person and for a fit person.



- (a) The heart rate of the unfit person decreased by 16%, 120 seconds after exercise.

Calculate the difference in percentage decrease in heart rate of these two people, 120 seconds after exercise.

Give your answer to an appropriate number of decimal places.

(3)

Answer

- (b) Calculate the heart rate of the unfit person at 5 minutes.

The change in heart rate of the unfit person is a linear relationship.
Use the equation for a straight line $y = mx + c$.

(3)

Answer

(c) Explain the difference in the time it would take for these two individuals to recover from the exercise.

(3)

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(d) Describe how heart rate is controlled in response to changes in the blood, following exercise.

(4)

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(Total for Question 6 = 13 marks)